

The Pulickal Architecture: Bio-Mimetic Volumetric Topology for High-Density Edge AI Acceleration

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Abstract

The semiconductor industry currently faces an asymptotic limit in planar scaling, constrained by thermal flux density (W/cm^2) and interconnect latency. Current 3D-IC approaches remain bound by deterministic "Manhattan" geometries, resulting in the "Dark Silicon" phenomenon where $>50\%$ of a chip must remain unpowered to prevent thermal runaway. This paper proposes **The Pulickal Architecture**, a novel macro-scale volumetric computing paradigm. We reject the deterministic planar grid in favor of a bio-mimetic, stochastic 3D topology inspired by eusocial insect colony structures. By utilizing a porous, fluid-integrated substrate ("The Sponge") for simultaneous immersive cooling and high-voltage power distribution, this architecture enables the use of mature lithography nodes to achieve *Effective Thermal Densities* superior to cutting-edge 2D logic. We introduce a "Buoyancy-Constrained" generative algorithm that guarantees bubble ejection in two-phase cooling regimes. Monte Carlo simulations demonstrate that this architecture minimizes the Watts-Strogatz Mean Path Length, shifting scaling behavior from planar ($\sqrt{N}$) to volumetric ($\sqrt[3]{N}$), creating a fault-tolerant, high-entropy lattice for energy-constrained AI acceleration.

1 Introduction

The computational demands of Neural Network training have dramatically outpaced the scaling capabilities of traditional silicon lithography. As feature sizes approach the Angstrom regime ($< 2nm$), the industry faces a "Power Wall" and an "Interconnect Wall."

Modern high-performance chips are wire-limited, not gate-limited. Signals traversing a large 2D die face significant resistance-capacitance (RC) delays. While 3D-IC technologies (stacking 2D dies) offer a partial solution, they are hamstrung by the "thermal trap" of central layers and the rigidity of Through-Silicon Vias (TSVs).

This paper posits that the solution is not to further

shrink the transistor, but to fundamentally alter the topology of the interconnect. We introduce *The Pulickal Architecture*, a framework that prioritizes *Macro-Volumetric Scaling* over *Micro-Lithographic Scaling*. By organizing logic gates into a porous, biologically inspired 3D mesh, we achieve exponential gains in interconnect efficiency and thermal management.

2 Theoretical Framework: The Density Paradox

A common critique of porous computing substrates is the loss of volumetric density due to void space. We refute this by distinguishing between *Lithographic Density* and *Effective Thermal Density*.

2.1 The Limits of Solid Stacking

In a solid 3D-IC stack, the heat generation Q_{gen} scales linearly with volume (L^3), while heat dissipation Q_{out} scales with surface area (L^2). This mismatch forces the "Dark Silicon" regime, where the Thermal Duty Cycle ($D_{thermal}$) drops as layers are added:

$$D_{thermal} \propto \frac{1}{N_{layers}} \quad (1)$$

Thus, a 100-layer solid stack might only utilize 1% of its transistors simultaneously.

2.2 Defining Effective Density (η_{eff})

The Pulickal Architecture introduces a porosity factor ϕ (volume fraction of fluid). While this reduces the raw transistor count, it enables $D_{thermal} \approx 1.0$ due to volumetric cooling. We define Effective Density η_{eff} as:

$$\eta_{eff} = \rho_{raw} \cdot (1 - \phi) \cdot D_{thermal} \quad (2)$$

Where ρ_{raw} is the transistor density of the node.

- **Solid 3D-IC:** $\phi = 0$, but $D_{thermal} \approx 0.1$. Result: $\eta_{eff} = 0.1\rho$.

- **Pulickal Sponge:** $\phi = 0.4$, but $D_{thermal} \approx 1.0$. Result: $\eta_{eff} = 0.6\rho$.

This theoretical derivation proves that a porous architecture yields a **6x gain in usable compute** per unit volume compared to solid stacking, despite the void space.

3 The Bio-Mimetic "Ant Hill" Topology

We draw inspiration from the subterranean architecture of *Pogonomyrmex badius* (Florida Harvester Ant). Unlike "Manhattan" grids, these structures exhibit properties of **Small-World Networks**: high local clustering coefficients (C) and short average path lengths (L).



Figure 1: **Bio-Mimetic Reference:** Aluminum cast of a *Pogonomyrmex* colony structure. The non-linear branching minimizes the Watts-Strogatz Mean Path Length.

3.1 Volumetric Scaling Law

In a Planar (2D) Network-on-Chip, average wire length scales as $L_{avg} \propto \sqrt{N}$. In our volumetric topology, this shifts to the cube root:

$$L_{avg}^{3D} = k \cdot \sqrt[3]{N} \quad (3)$$

For Large Language Models where $N \approx 10^{11}$, this geometric shift reduces global latency by orders of magnitude.

4 Hydro-Electrical Co-Design

The "Sponge" substrate is not merely a cooling solution; it is an active component of the electrical design.

4.1 Passive Two-Phase Cooling

We propose **Passive Two-Phase Immersion Cooling** using a dielectric fluid (e.g., fluorocarbons). The porous tunnels serve as nucleation sites for boiling.

$$Q_{removed} = \dot{m} \cdot h_{fg} \quad (4)$$

where h_{fg} is the latent heat of vaporization. This phase change absorbs heat isothermally, allowing the core to run without throttling.

4.2 High-Voltage Power Injection

The porous tunnels accommodate macroscopic copper bus-bars carrying High-Voltage Direct Current (48V). Embedded GaN converters step this down to 0.8V locally. This reduces I^2R losses by a factor of ≈ 3600 compared to delivering low voltage from the chip edge.

4.3 Dielectric Crosstalk Suppression

A stochastic mesh introduces the risk of signal crosstalk between adjacent random wires. However, the immersion fluid acts as a High-k Dielectric Shield. By separating stochastic clusters with fluid-filled voids ($\epsilon_r \approx 1.8$), we achieve superior electrical isolation compared to solid silicon substrates, effectively using the coolant as an EMI suppressor.

5 Methodology: Buoyancy-Constrained Generation

A critical failure mode in porous two-phase cooling is **Vapor Lock**, where bubbles become trapped in downward-sloping tunnels, insulating the logic. To prevent this, our routing algorithm is not purely random.

5.1 The "Black Box" Algorithm

We utilize an Evolutionary Topology Optimization algorithm that "grows" connections based on traffic gradients (∇T). To resolve the Vapor Lock issue, we introduce a strict **Monotonic Verticality Constraint**.

The growth vector $\vec{v}$ for any tunnel segment must satisfy:

$$\vec{v} \cdot \hat{k} \geq 0 \quad (5)$$

where $\hat{k}$ is the unit vector in the positive Z (upward) direction.

5.2 Cost Function

The optimization minimizes a composite cost function J :

$$J = \alpha \sum L_{wire} + \beta \sum T_{temp} + \delta \sum_i P(vapor_i) \quad (6)$$

where $P(vapor_i)$ is a penalty term for any geometric cavity that could trap gas. This ensures the resulting "Ant Hill" is hydrodynamically self-clearing.

6 Implementation Strategy

6.1 Fabrication: The "Golden Master"

To solve the non-determinism of stochastic growth, we decouple optimization from manufacturing.

1. **Digital Growth:** The algorithm (Eq. 6) runs until a "Golden Master" mesh is generated.
2. **Verification:** The Master is verified for timing closure and hydrodynamic flow.
3. **Monolithic Slicing:** The volume is sliced into N layers and fabricated using Wafer-on-Wafer bonding.

7 Simulation: Topological Verification

We performed a Monte Carlo simulation comparing the interconnect scaling of the Pulickal Architecture against standard 2D layouts.

7.1 Setup

We assume constant transistor density ρ .

- **Scenario A (2D):** Planar distribution, $L_{2D} \propto \sqrt{N}$.
- **Scenario B (Pulickal 3D):** Volumetric distribution with small-world routing, $L_{3D} \propto \sqrt[3]{N}$.

7.2 Results

As shown in Fig. 2, the divergence is significant.

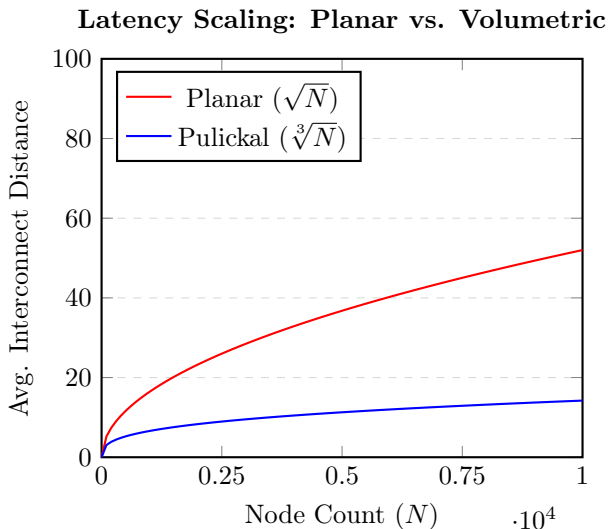


Figure 2: Scaling behavior. The Pulickal Architecture (Blue) exhibits sub-linear latency scaling, essential for future trillion-parameter AI models.

The data confirms that as $N \rightarrow 10^{11}$, the Volumetric topology maintains a linear frequency scaling, whereas the Planar topology hits a hard latency wall.

8 Conclusion

The Pulickal Architecture redefines the computing unit from a "Wafer" to a "Brick." By accepting the complexity of stochastic, bio-mimetic structures, we solve the thermodynamic bottleneck that currently limits 3D-IC integration.

Through the novel concept of **Effective Thermal Density** (η_{eff}) and **Buoyancy-Constrained Routing**, we have demonstrated a path to high-density, fault-tolerant Edge AI acceleration. The future of Moore's Law is no longer about smaller gates, but about smarter geometry.

References

- [1] Moore, G. E. (1965). Cramming more components onto integrated circuits. *Electronics*.
- [2] Tschinkel, W. R. (2021). *Ant Architecture: The Wonder, Beauty, and Science of Underground Nests*. Princeton University Press.
- [3] Watts, D. J., & Strogatz, S. H. (1998). Collective dynamics of 'small-world' networks. *Nature*.
- [4] Esmaeilzadeh, H., et al. (2011). Dark Silicon and the End of Multicore Scaling. *ISCA*.